



# 2024 Privacy and Public Policy Conference

## OPTIMAL RESOLUTION OF A DATA SHARING TRILEMMA: STATISTICAL POWER, SAMPLE COMPLEXITY, AND PRIVACY BUDGET

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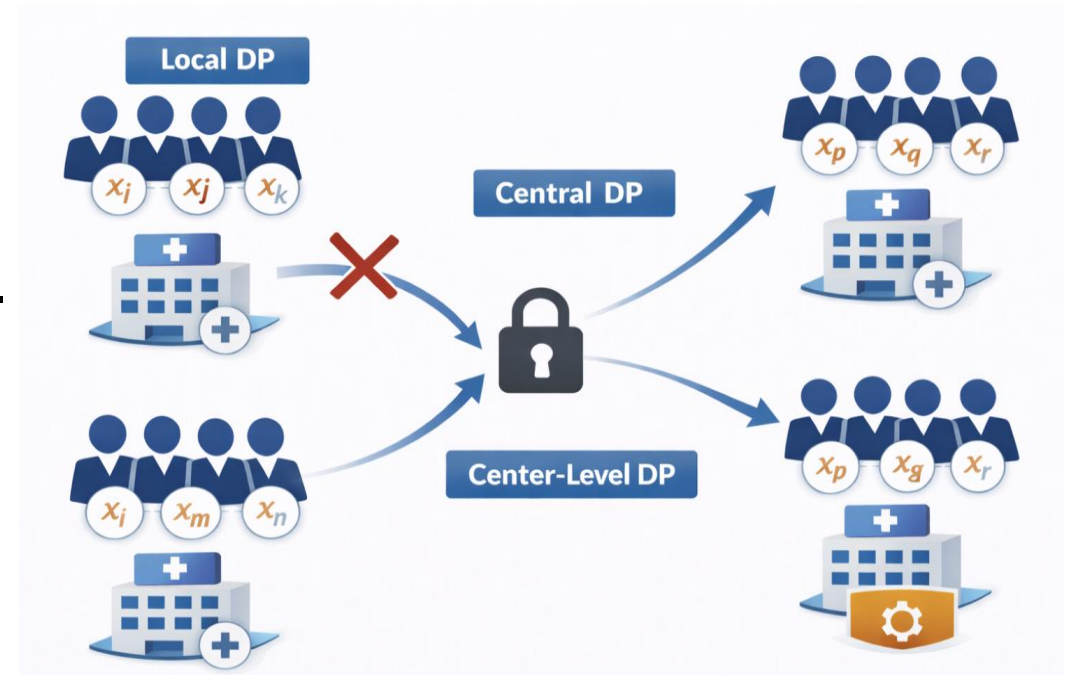


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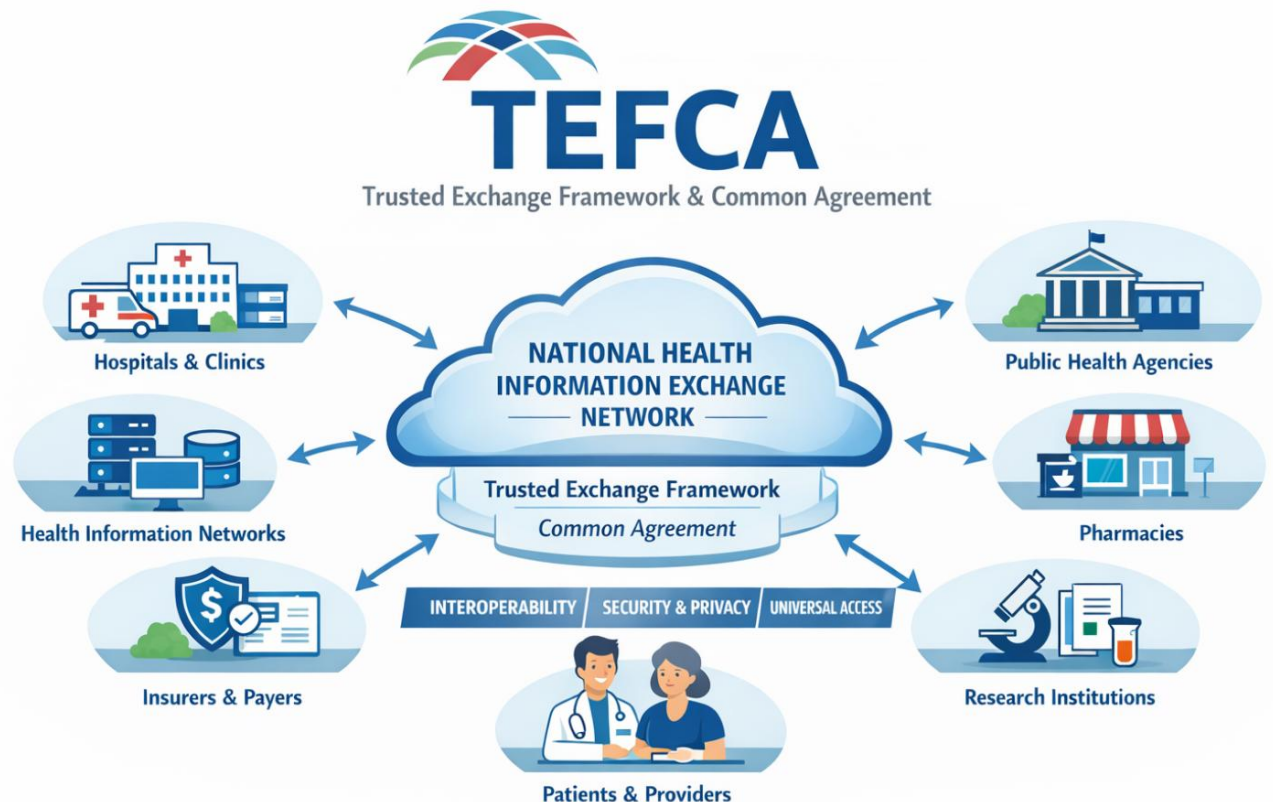
# Motivation

- In many real-world collaborations, *institutions*—not individuals—decide whether to share data.
- Existing privacy models focus on two extremes:
  - **Local DP**: protects each individual record
  - **Central DP**: assumes a fully trusted aggregator
- In practice, institutions are often willing to share *summary statistics*, but must protect the privacy of their *internal datasets*.



# A Real-World Example: TEFCA

- **TEFCA** establishes a national framework for health data exchange in the U.S.
- Under TEFCA, **institutions—not individuals—control data sharing**.
- Data are shared primarily as **aggregated or summary statistics**, under strict governance.
- Institutions must protect the privacy of their **internal datasets** while participating.



# From Practice to Modeling — Three Privacy Mechanisms



- Local Privacy (Individual-Level DP)**

Each individual perturbs their own data before sharing.

→ Strong individual protection, but often low statistical power.

- Center-Level Privacy (Institution-Level DP)**

Each institution shares privatized **summary statistics**, protecting its internal dataset.

→ Matches real-world collaborations such as TEFCA.

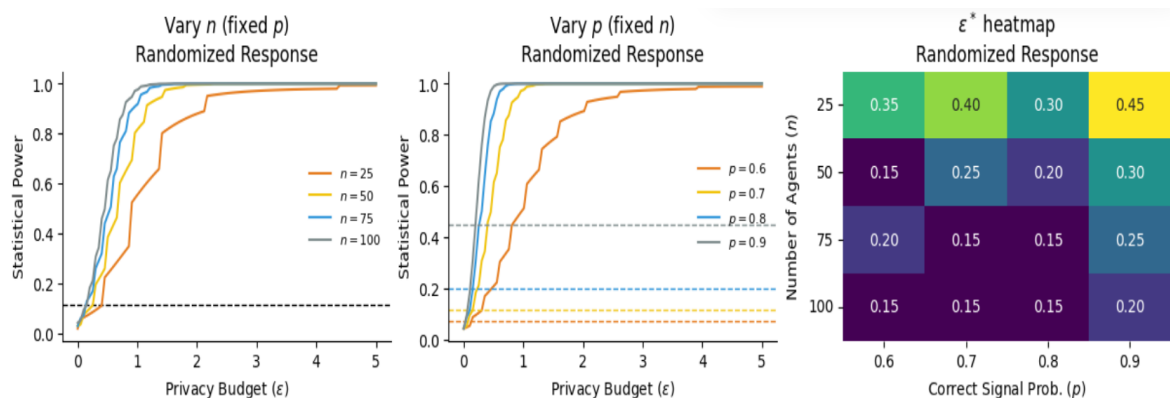
- Central Privacy (Aggregator-Level DP)**

Institutions share exact summaries with a trusted aggregator, which adds noise only at release.

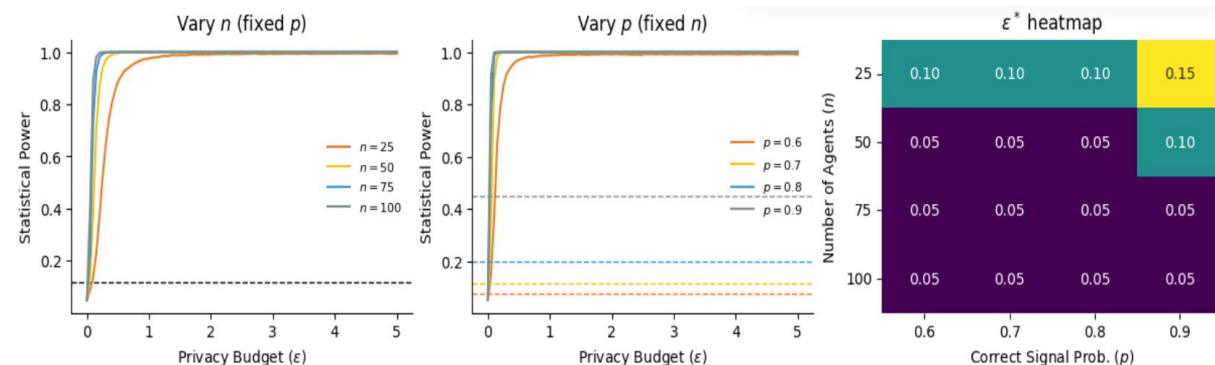
→ High utility, but requires strong trust assumptions.

# Result

- Local Privacy Mechanism



- Central Privacy Mechanism



- Center-Level Privacy Mechanism

